

Hidden Networks: Automated road and footpath extraction using cascaded CNNs in Nakivale, Uganda

By Zachary Mondschein, October 17, 2025

What are you trying to do:

I aim to assess the application of a deep learning computer algorithm to automatically map roads and footpaths inside of the large refugee settlement of Nakivale, Uganda. My research questions are threefold: (1) Can a lightweight CNN architecture, originally developed and benchmarked on urban road systems, be adapted to humanitarian settings where infrastructure is informal and sometimes ephemeral? (2) Can we train a model using annotations for roads and footpaths that are adapted from OpenStreetMap data, despite incomplete and uneven coverage in the training dataset? (3) Does adding small amounts of high-quality annotations significantly improve model accuracy?

How is it done today, and what are the limits of current practice?

In Uganda, infrastructure mapping is facilitated by the Humanitarian OpenStreetMap Team (HOT OSM) and local collaborators MapUganda using manual digitization techniques directly in the field. This approach is preferred for rural systems, where diverse and informal roadways are difficult to identify using earth observation, and because very high-resolution imagery is not always accessible to planners due to resource constraints. However, in refugee settlements, where population influx can be large and where infrastructure can evolve quickly, manual approaches are unable to keep up with the pace of spatial change. As a result, existing road and footpath maps are frequently incomplete or outdated.

What is new in your approach and why do you think it will be successful?

I will build a modern deep-learning model specifically for humanitarian landscapes using 0.5 m Maxar imagery and annotations from a combination of existing OSM polyline layers (adapted to binary pixel masks), and new hand-labeled examples of missing roadways. The method that I am adapting also employs a two-step approach to encourage the network to produce topologically corrected road maps.

Who cares? If you are successful, what difference will it make?

Roads and footpaths enable planners and researchers to better understand the connectivity and lifeways of people who reside in refugee settlements. In the Conflict Ecology lab, we intend to use updated infrastructure maps to estimate travel times within settlements (for agriculture, school, and resource gathering activities), a main risk factor for exposure to gender-based violence.

What are the risks?

The model may not result in a substantial improvement in infrastructure mapping over manual methods. Outputs may not be topologically aware and may require substantial postprocessing, which decreases the added value of a modelling approach.

How much will it cost?

The financial costs of this project are minimal. It will require compute time and storage on university HPC infrastructure. Manual labelling will be provided by university students (~20 hours). Costly VHR Maxar imagery was previously acquired for other projects.

How long will it take? 2-4 months

What are the mid-term and final “exams” to check for success?

The midterm and final exams for this project, in the best-case scenario correspond to the midterm and final presentations for this course. A clear project development plan will be presented during the midterm, and preliminary results will be presented during the final presentation. We will evaluate the model based on machine learning prediction metrics and visual interpretation of results.